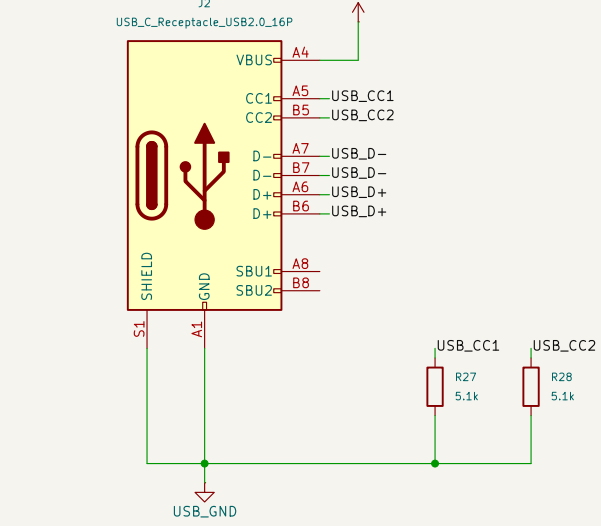


AK4490 USB DAC

SIGNAL FLOW USB INPUT > ISOLATION > XMOS / I2S > DAC > ANALOG FILTER / LINE OUTPUT

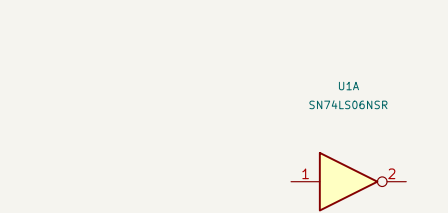
01 USB-C INPUT

PC / USB bus power / USB_GND



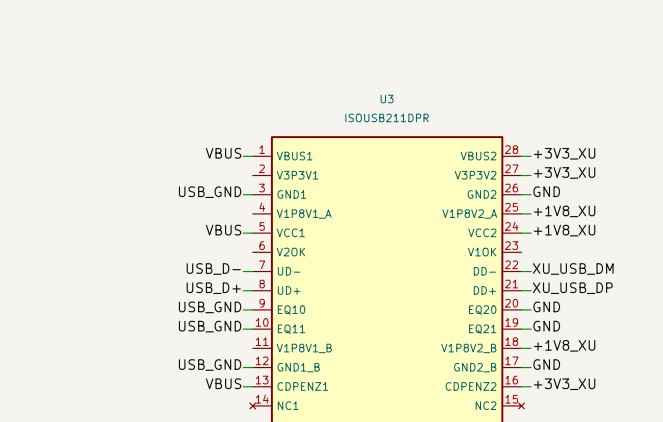
USB signals connect by matching net labels.

AUXILIARY LOGIC



02 USB ISOLATION

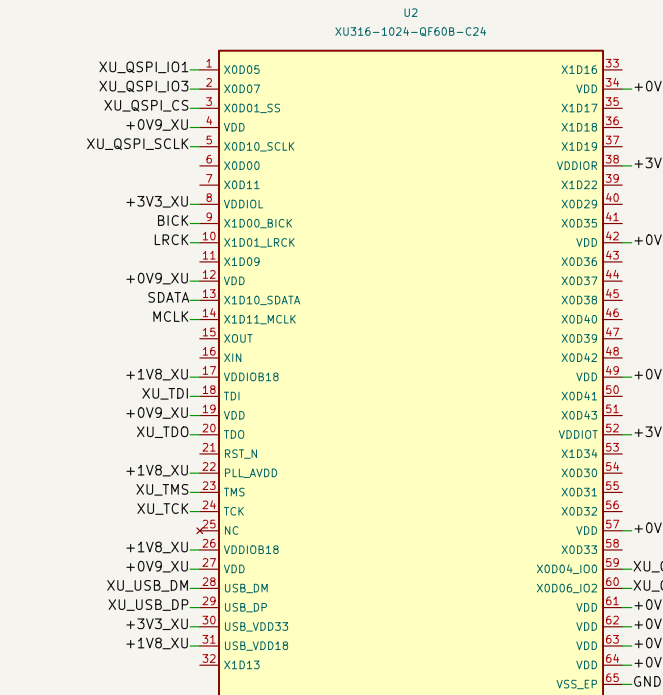
ISOU8211 | USB_GND <=> main GND



Isolation domains are retained.

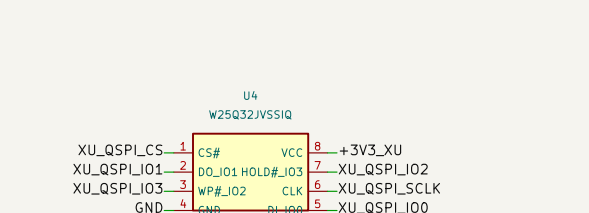
03 XMOS / USB AUDIO

XU316 | I2S / QSPI boot / clock / debug

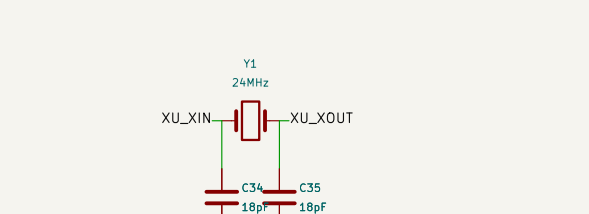


I2S GPIO assignments are provisional.

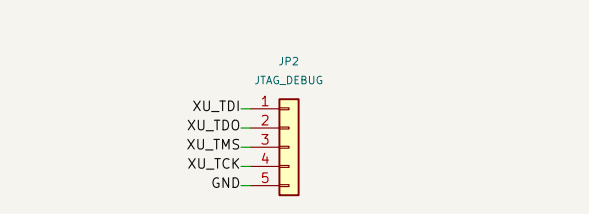
QSPI BOOT FLASH



24 MHz CLOCK

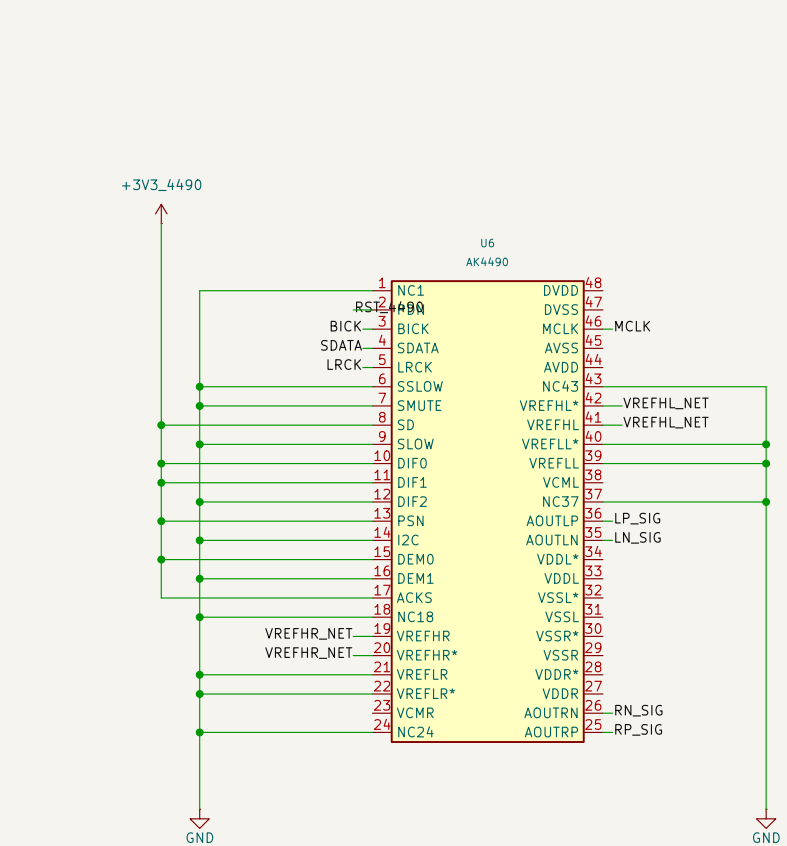


JTAG DEBUG

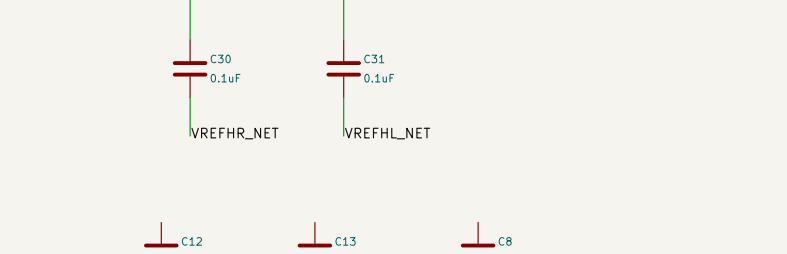


04 DAC / REFERENCES

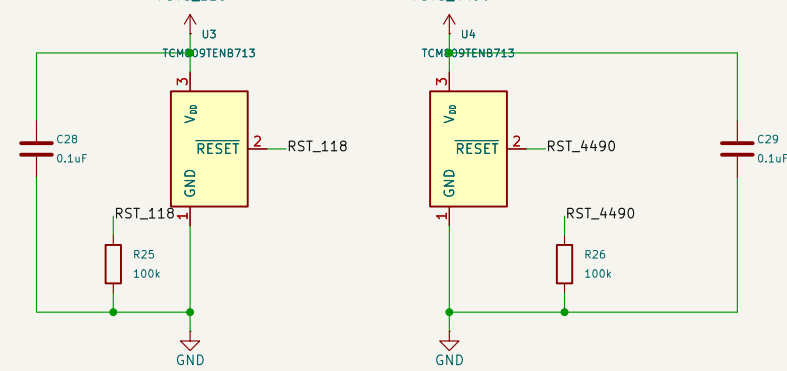
AK4490 | digital control / analog reference



REFERENCE BYPASS



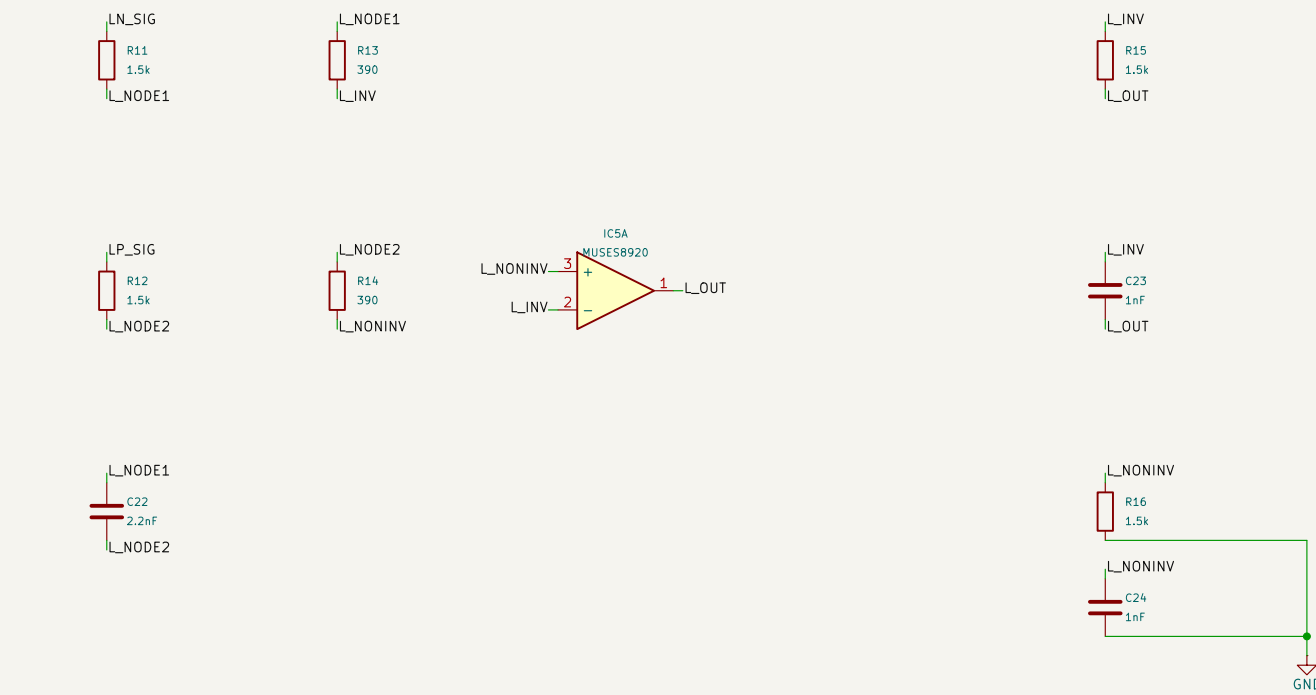
RESET SUPERVISORS



U3 / U4 reference duplicates exist in the source design.

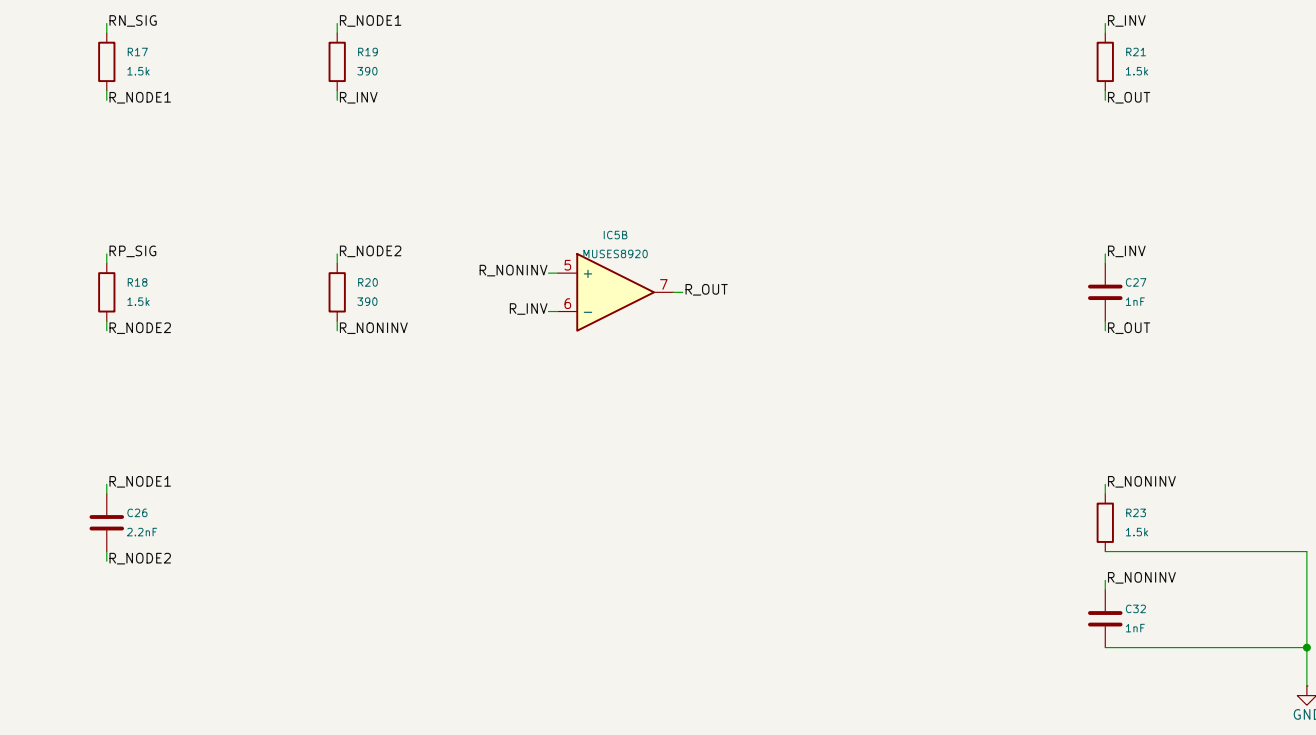
05 LEFT ANALOG OUTPUT

Differential input > filter / MUSES8920 > LINE_OUT_L



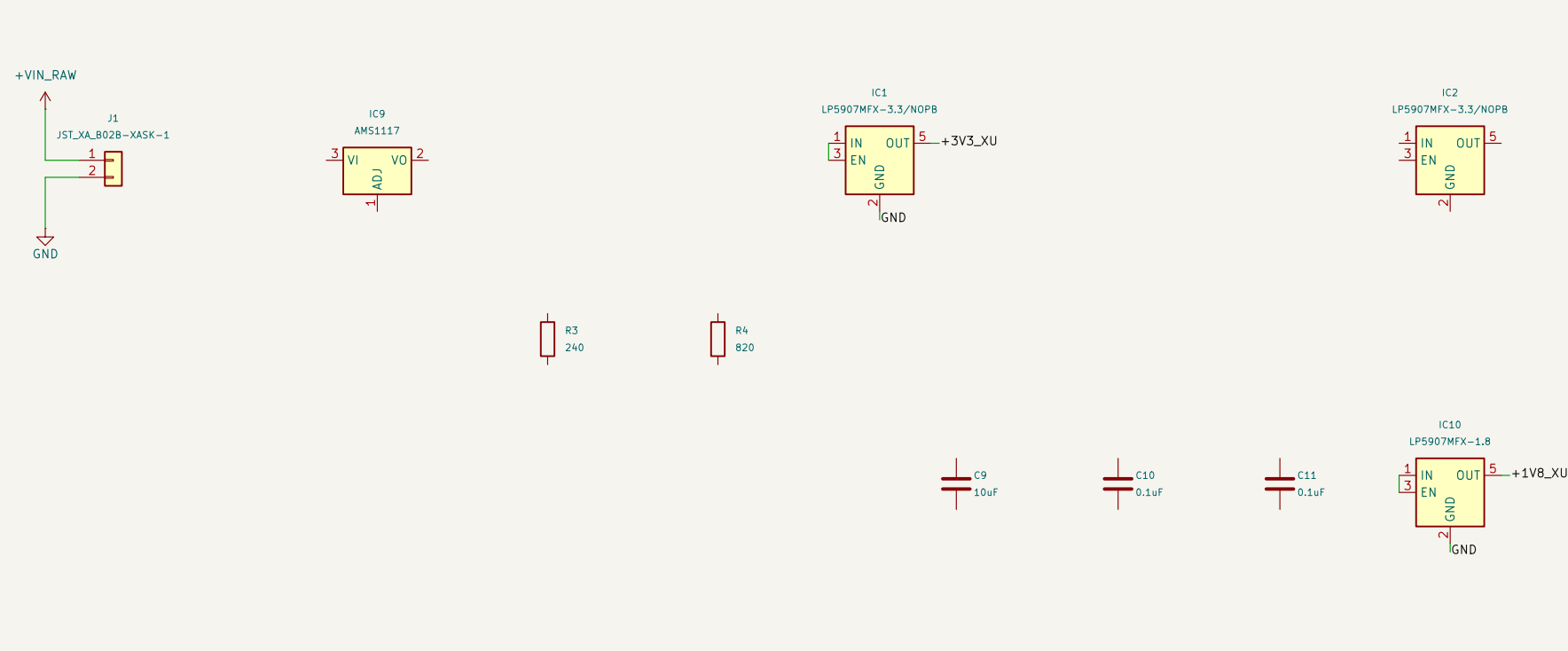
06 RIGHT ANALOG OUTPUT

Differential input > filter / MUSES8920 > LINE_OUT_R

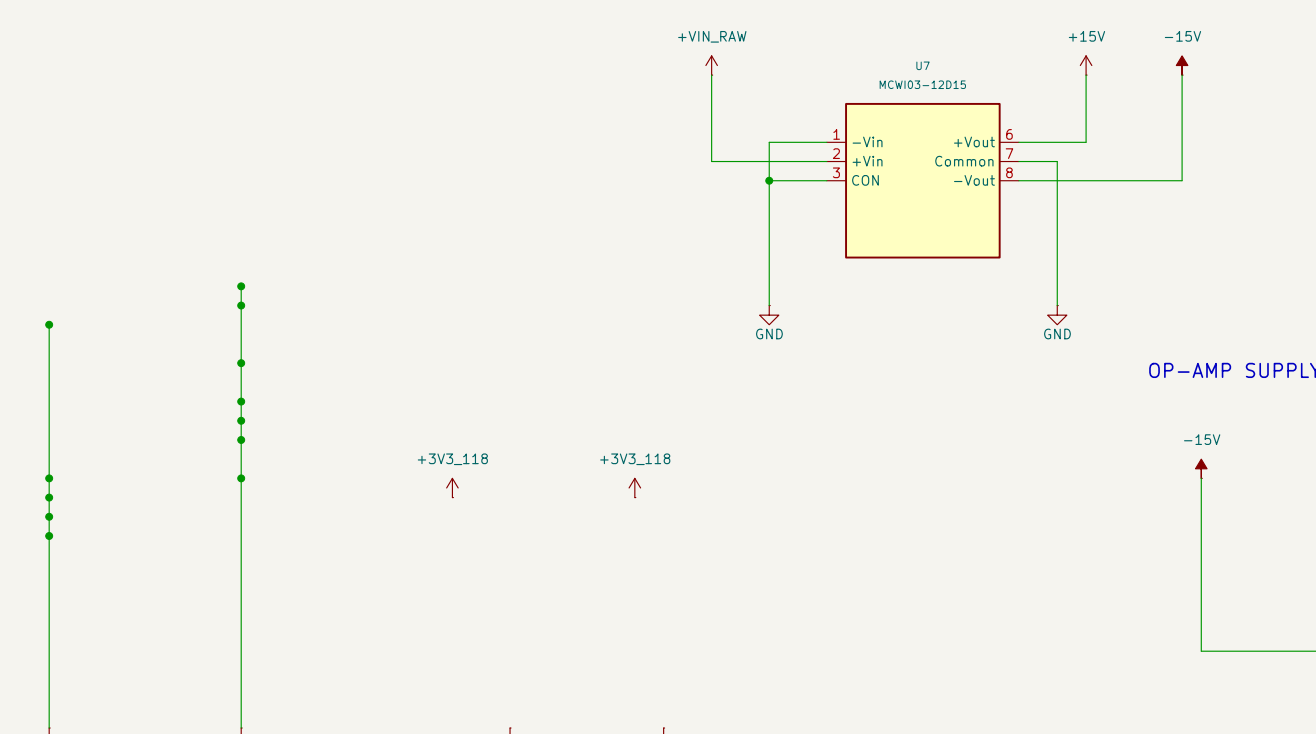


07 POWER SUPPLIES

DC input / digital rails / DAC rails / analog +/-15 V



Existing unconnected pins are retained.



Layout cleanup only. Existing incomplete connections retained.
USB-C > ISOU8211 > XMOS XU316 > AK4490 > stereo line output

Sheet: /
File: AK4490DAC.kicad_sch
Title: **AK4490 USB DAC - functional layout**
Size: User Date:
Kicad E.D.A. 10.0.3 Id: 1/1